AI Family Concept Map

1. ARTIFICIAL INTELLIGENCE (AI)

The broad field of making machines smart and able to perform tasks that normally require human intelligence.

Includes:
- Machine Learning
- Deep Learning
- Generative AI
- Large Language Models

and more...

Real-world Example

Siri / Google Assistant

(Understands voice, answers questions, performs tasks)

AI is the umbrella.
ML learns from data.
DL uses deep neural networks.
GenAI creates new content.
LLMs understand & generate human-like text.

2. MACHINE LEARNING (ML)

A subset of AI that enables machines to learn from data and improve performance on tasks without being explicitly programmed.

Real-world Example

Email Spam Detection

(Learns from emails and detects spam)

3. DEEP LEARNING (DL)

A subset of ML that uses artificial neural networks with many layers to learn from large amounts of data and find complex patterns.

Real-world Example

Image Recognition

(Identifies objects, faces, scenes in images)

HOW THEY RELATE (HIERARCHY)

ARTIFICIAL INTELLIGENCE (AI)
↓
MACHINE LEARNING (ML)
↓
DEEP LEARNING (DL)
↓
GENERATIVE AI (GenAI)
↓
LARGE LANGUAGE MODELS (LLMs)

Each level is a subset of the above level.

$AI \supset ML \supset DL \supset GenAI \supset LLMs$

4. GENERATIVE AI (GenAI)

A subset of DL that can generate new content such as text, images, audio, code, etc., based on patterns learned from existing data.

Real-world Example

AI Image Generation (DALL·E, Midjourney)

5. LARGE LANGUAGE MODELS (LLMs)

A type of Generative AI model trained on huge text datasets to understand and generate human-like text.

Real-world Example

ChatGPT / Google Gemini

(Text generation, Q&A, translation, summarization)

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